计算机组成与设计 · 计组期末考试A

1. What is the range of exponent of IEEE 745 single precision?

单选题 (3.0 分) （难易度:中）

A. 1~254

B. -128~126

C. -126~127

D. -127~128

正确答案：C

答案解释：暂无

2. The SRAMs are basically used as _____

单选题 (3.0 分) （难易度:中）

A. register

B. cache

C. main memory

D. disk

正确答案：B

答案解释：暂无

3. What is the decimal product of the binary number 1.00×2^{-1} and -1.11×2^{-2}?

单选题 (3.0 分) （难易度:中）

A. -0.4375

B. 0.0625

C. -0.21875

D. -0.0625

正确答案：C

答案解释：暂无

4. Consider the following C code:

```
typedef unsigned char *pointer; // sizeof(unsigned char) = 1 byte
void show_bytes(pointer start, size_t len) {
for (int i = 0; i < len; i++)
printf("0x%x\n", start[i]);
}
int main() {
int a = 0x11223344;
show_bytes((pointer) &a, sizeof(int));
}
```

If this C code runs on a little-endian machine, what will we get on the **third** line of the terminal output? ()

单选题 (3.0 分) （难易度:中）

A. 0x11

B. 0x22

C. 0x33

D. 0x44

正确答案：B

答案解释：暂无

5. The reason for the implementation of the cache memory is _____.

单选题 (3.0 分) （难易度:中）

A. to increase the internal memory of the system

B. the difference in speeds of operation of the processor and memory

C. to reduce the memory access and cycle time

D. all of the mentioned

正确答案：B

答案解释：暂无

6. Which of the following allows simultaneous write and read operations?

单选题 (3.0 分) （难易度:中）

A. ROM

B. EROM

C. RAM

D. None of the above

正确答案：C

答案解释：暂无

7. The copy-back protocol is used _____.

单选题 (3.0 分) （难易度:中）

A. to copy the contents of the memory onto the cache

B. to update the contents of the memory from the cache

C. to remove the contents of the cache and push it on to the memory

D. none of the mentioned

正确答案：B

答案解释：暂无

8. Consider a virtual memory system with 32 bits virtual byte address, 4KiB/page, 32 bits each entry. The physical memory is 512MiB. Then, the total size of a single-level page table needs _____

单选题 (3.0 分) （难易度:中）

A. 1MiB

B. 4MiB

C. 8MiB

D. 16MiB

4byte virtual address size=2^32bits page size=2^12

virtual page number=VPN=2^32/2^12=2^20

page table size=page number*page size=2^20*4byte=4MB

正确答案：B

答案解释：暂无

9. The function of assembler is _____.

单选题 (3.0 分) （难易度:中）

A. Transforming high level language to binary language

B. Transforming binary language to high level language

C. Transforming assembly language to machine code

D. Transforming high level language to assembly language

正确答案：C

答案解释：暂无

10. The temporal aspect of the locality of reference means _____.

单选题 (3.0 分) （难易度:中）

A. That the recently executed instruction won’t be executed soon

B. That the recently executed instruction is temporarily not referenced

C. That the recently executed instruction will be executed soon again

D. None of the mentioned

正确答案：C

答案解释：暂无

11. A given application written runs 15 seconds on a desktop processor. A new compiler is released that requires only 0.6 as many instructions as the old compiler. Unfortunately, it increases the CPI by 1.1. How fast can we expect the application to run using this new compiler?

单选题 (3.0 分) （难易度:中）

A. 8.2sec

B. 9.9sec

C. 27.5sec

D. 22.7sec

正确答案：B

答案解释：暂无

12. Given the following RISC-V assembly code (and assuming all registers start at 0):

```
        addi   t1, x0, 10
        add    t2, t1, t1
repeat: addi   t2, t2, -4
        add    t3, t2, t2
        addi   t1, t1, -2
        bne    x0, t1, repeat
```

What is the final value of register t3?

单选题 (3.0 分) （难易度:中）

A. 0

B. -4

C. 8

D. 4

正确答案：A

答案解释：暂无

13. A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because_____

单选题 (3.0 分) （难易度:中）

A. it reduces the memory access time to read or write a memory location

B. it helps to reduce the size of page table needed to implement the virtual address space of a process

C. it is required by the translation lookaside buffer

D. it helps to reduce the number of page faults in page replacement algorithms

正确答案：B

答案解释：暂无

block=32B/8B=4,entry=4/2=2,index=1,offset=3

14. You have a two-way set-associative cache with 8B blocks and a total size of 32B. Suppose use a least-recently used replacement policy and begin from power on. Given this sequence of byte addressed accesses, what is the hit rate? (_____)
    0, 4, 128, 24, 224, 88, 134, 92

单选题 (3.0 分) （难易度:中）

A. 12.5%

B. 25.0%

C. 37.5%

D. 50.0%

正确答案：C

答案解释：暂无

15. Which one is not one of the five classic components of a computer?

单选题 (3.0 分) （难易度:中）

A. Input

B. Bus

C. Memory

D. Output

正确答案：B

答案解释：暂无

16. What is the range of 32-bit instructions that can be reached from the current PC using a UJ-Format jump instruction?

结果要/4

单选题 (3.0 分) （难易度:中）

A. $[-2^{21}, 2^{21} - 1]$

B. $[-2^{20}, 2^{20} - 1]$

C. $[-2^{19}, 2^{19} - 1]$

D. $[-2^{18}, 2^{18} - 1]$

正确答案：D

答案解释：暂无

17. Calculate AMAT (Average Memory Access Time) for a machine with the following specs: L1 cache with hit time = 1 cycle and miss rate = 5%, L2 cache with hit time = 5 cycles, miss rate = 15% and miss penalty = 200 cycles.

单选题 (3.0 分) （难易度:中）

A. 2.75 cycles

B. 2 cycles

C. 1.665 cycles

D. None of the above.

正确答案：A

答案解释：暂无

18. Assume a 5-stage pipelined RISC-V CPU with no forwarding. How many stalls would there need to be in order to fix all data hazards (assuming that we can read and write to the RegFile on the same cycle)?

```
sub t1, s0, s1
or s0, t0, t1
sw s1, 100(s0)
add s2, s0, s2
lw t1, 104(s0)
```

单选题 (3.0 分) （难易度:中）

A. 2

B. 3

C. 4

D. 5

正确答案：C

答案解释：暂无

19. The bit width of PC is determined by _____.

单选题 (3.0 分) （难易度:中）

A. Memory word length

B. Memory capacity

C. Instruction word length

D. Bit width of general-purpose registers

正确答案：B

答案解释：暂无

20. The fastest data access is provided using _____.

单选题 (3.0 分) （难易度:中）

A. Caches

B. DRAMs

C. SRAMs

D. Registers

正确答案：D

答案解释：暂无

21. The LRU can be improved by providing a little randomness in the access.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

22. The associative mapping is costlier than direct mapping.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

23. In the memory hierarchy, as the speed of operation increases the memory size also increases.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：B

答案解释：暂无

24. The pipeline bubbling is a method used to prevent data hazard and control hazards.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：B

答案解释：暂无

25. Virtual memory allows a single program to expand its address space beyond the limits of main memory.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

26. Cache block size (B) can affect both miss rate and miss latency.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

27. There is no way to reduce compulsory misses.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：B

答案解释：暂无

28. The CPI of superscalar processors can be less than one.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

29. The higher the memory bandwidth, the larger the cache block.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：A

答案解释：暂无

30. The page table is stored in the disk.

判断题 (1.0 分) （难易度:中）

A. True

B. False

正确答案：B

答案解释：暂无

31. **Assembly and Pipeline (10 points)**

Consider the following sequence of instructions:

```
loop: add t0, t1, t2
lw t3, 10(t0)
lw t4, 14(t0)
sub t5, t4, t3
sw t5, 18(t0)
addi t2, t2, 4
slti t6, t2, 200
bne t6, x0, loop
```

Assume each datapath stage requires the following amount of time to complete:
Instruction fetch (IF): 30 ns
Instruction decode (ID): 20 ns
Execute / address calculation (EX): 25 ns
Memory access (MEM): 30 ns
Register write back (WB): 20 ns

综合题 (10.0 分) （难易度:中）

(1) How long will a single iteration of this loop take in a single-cycle datapath? (number only) 1 ns

填空题 (2.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

正确答案：
① 1000

答案解释： 暂无

(2) If we assume ideal pipelining (i.e., no hazards and therefore no stalls), how long will one loop iteration take in a pipelined datapath?(number only) 1 ns

填空题 (2.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

正确答案：
① 360

答案解释： 暂无

(3) If we now assume a pipelined datapath with forwarding, how many data hazard(s) cannot be solved with forwarding? (number only) 1

填空题 (3.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

正确答案：

① 1

答案解释：暂无

(4) If we now assume a pipelined datapath with forwarding, how long will one iteration take?(number only)__1__ns

填空题 (3.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

正确答案：
① 390

答案解释：暂无

32. **CACHE**

You are trying to reverse-engineer the characteristics of a cache in a system, so that you can design a more efficient, machine-specific implementation of an algorithm you are working on. To do so, you have come up with two sequences of memory accesses to various bytes in the system in an attempt to determine the following four cache characteristics:
Cache block size (16, 32, 64, or 128 B).
Cache associativity (1-, 2-, 4-, or 8-way).
Cache size (4 or 8 KiB).
Cache replacement policy (LRU).
The only statistic that you can collect on this system is cache hit rate after performing each sequence of memory accesses. Here is what you observe:

	Addresses Accessed (Oldest → Youngest)								Hit Rate
1.	0	23	128	73	8192	255	16384	196	1/2
2.	127	4096	8192	32768	196	16384	0	512	3/8

Assume that the cache is initially empty at the beginning of the first sequence, but not at the beginning of the second sequence. The sequences are executed back-to-back, i.e.,no other accesses take place in between the two sequences. Thus, at the beginning of the second sequence, the contents are the same as at the end of the first sequence.

Based on what you observe, what are the following characteristics of the cache? Choose your answer.

综合题 (10.0 分) （难易度:中）

(1) Cache block size (16, 32, 64, or 128 B)?

单选题 (3.0 分) （难易度:中）

A. 16B

B. 32B

C. 64B

D. 128B

正确答案：D

答案解释：暂无

(2) Cache associativity (1-, 2-, 4-, or 8-way)?

单选题 (3.0 分) （难易度:中）

A. 1-way

B. 2-way

C. 4-way

D. 8-way

正确答案：C

答案解释：暂无

(3) To identify the cache size, you execute the following sequence right after sequence 2 (i.e., the contents are the same as at the end of the second sequence) and measure the cache hit rate:
Addresses Accessed (Oldest → Youngest): 8192 → X → Y
Which addresses should you use for X and Y?
X:_____

单选题 (2.0 分) （难易度:中）

A. 1024

B. 2048

C. 4096

D. None of above

正确答案：A

答案解释：暂无

(4) Y:_____

单选题 (2.0 分) （难易度:中）

A. 8192

B. 16384

C. 32768

D. None of above

正确答案：C

答案解释：暂无

33. VIRTUAL MEMORY

Consider a processor that includes a 40-bit virtual address, an MMU that supports 4096 (2^{12}) bytes per page, 2^{32} bytes of physical memory, and a large Flash memory that serves as a disk. The MMU and the page fault handler implement an LRU replacement strategy.

综合题 (10.0 分) （难易度:中）

page size=2^12,cache size=2^40,page number=2^28,offset=12

(1) Please calculate the following parameters relating to the size of the page table. You may assume each page entry contains a valid bit and a dirty bit.(2^{10} = 2^10)
Number of entries in the page table:__1__ 2^28=256MB
Size of page table entry (in bits):__2__ physical page number=2^32/2^12=2^20
20+1+1=22

填空题 (2.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

①

②

正确答案：
① 2^28
② 22

答案解释：暂无

(2) A program running on the processor is halted right before executing the following instruction located at address 0x056C:

```
lw    x3, 0(x5)   // x5 = 0x2800
sw    x31,0(x6)   // x6 = 0x4200
```

0x056C+4=0x0570

The first 8 locations of the page table, just before executing this test program, are shown below; the least-recently-used page (“LRU”) and next least-recently-used page (“next LRU”) are as indicated. If pages must be brought in from disk, increment the next largest page number.

Page Table

	VPN	D	V	PPN
	00	1	1	0x7
	01	0	1	0x5
	02	0	1	0x3
LRU→	03	1	1	0x1
	04	--	0	--
	05	0	1	0x0
	06	0	1	0x2
Next LRU→	07	0	1	0x6

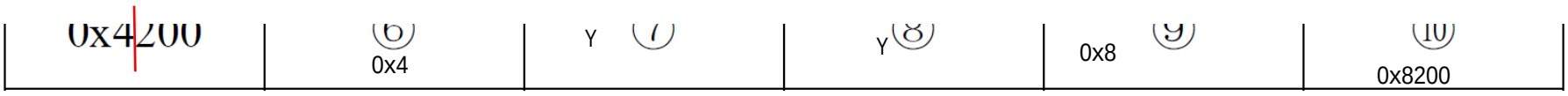
This processor also has a 4 element, fully associative, Translation Lookaside Buffer (TLB) that caches translations from VPN to PPN.

TLB

	Tag	D	V	PPN
LRU→	0x3	1	1	0x1
	0x2	0	1	0x3
	0x6	0	1	0x2
Next LRU→	0x1	1	1	0x5

For each virtual address in the chart below, please indicate the VPN, whether or not the access results in a TLB Miss, whether or not the access results in a page fault, the PPN, and the physical address. Please write all numerical values in hexadecimal.

Virtual Address	VPN	TLB Miss? (Y/N)	Page Fault? (Y/N)	PPN	Physical Address
0x2800	① 0x2	N ②	N ③	0x3 ④	⑤ 0x3800



填空题 (5.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

①

②

③

④

⑤

⑥

⑦

⑧

⑨

⑩

正确答案:
① 0x2 2
② N
③ N
④ 0x3 3
⑤ 0x3800 3800
⑥ 0x4 4
⑦ Y
⑧ Y
⑨ 0x8 8
⑩ 0x8200 8200

答案解释: 暂无

(3) What is the physical address of the **sw** instruction?
Physical address of sw instruction:0x__1__

填空题 (3.0 分) （难易度:中） (请按题目中的空缺顺序依次填写答案)

正确答案:
① 7570

0x056C+4=0x0570
VPN=0x0,TLB miss,PT hit,PPN=0x7,PA=0x7570

答案解释: 暂无